

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science and Engineering

Branch:- CSE

Semester – 2nd

Section- All sections

Mid-Term Examination March – 2026

Subject Name/ Code:- Digital Logic Design- CSE-2003

Max. Marks: 20

Time Allotted: 90 min

NOTE: 1. Section A is compulsory. Attempt Section B or C, anyone.

2. Mark answers number clearly.

3. Attempt All Questions in sequence and at one place.

S. No	Question	Marks
Section A		
1.	(a) Simplify the following Boolean function using Quine Mcclausky/Tabulation method. $F(A, B, C, D, E) = \sum m(0, 1, 4, 5, 16, 17, 21, 25, 29)$	5
	(b) A smart security and monitoring system in a high-tech facility uses five sensors represented by the variables A, B, C, D, and E. Each sensor outputs 1 when a specific condition is detected and 0 otherwise. The system activates an emergency alert ($F = 1$) when certain combinations of sensor signals occur. Based on the system design, the alert must be triggered for the following sensor combinations: $F(A, B, C, D, E) = \sum m(0, 2, 3, 4, 5, 6, 7, 11, 15, 16, 18, 19, 23, 27, 31)$ However, for some sensor states the output does not affect system operation and can be treated as don't-care conditions: $d(1, 8, 9, 25)$ As a digital system designer, your task is to optimize the logic used in the controller. (i) Use the Karnaugh Map (K-map) method to simplify the Boolean function for the alert signal F. (ii) Implement the simplified function using NOR gates only, since the hardware platform supports NOR logic modules.	5
	(c) Design a circuit using NAND gates only with five inputs and one output where the output is 1 if the input is divisible by 3 or 7.	5
Section B		
2.	(a) A manufacturing plant uses two tanks to store certain liquid chemicals that are required in a manufacturing process. Each tank has a sensor that detects when the chemical level drops to 25% of full. The sensors produce a HIGH level	3

(Please Turn Over)



		when the tank is more than one-quarter full. When the volume of chemicals in a tank drops to one-quarter full, the sensor puts out a LOW level. It is required that a single green light emitting diode on a indicator panel show when both tanks are more than one-quarter full. Design the truth table and find out the Boolean expression.	
	(b)	Represent given decimal number $(97)_{10}$ in Excess-3, Gray code and 8-4-2-1 code. State which one is weighted coding technique?	2
Section C			
3.	(a)	An automated irrigation system in a smart farm uses four inputs A, B, C, and D representing different environmental conditions such as soil moisture, temperature level, sunlight intensity, and rainfall detection. The system generates an output F to activate the water pump when specific combinations of these conditions occur. The pump should turn ON for the following input combinations: $F(A, B, C, D) = \Sigma(0, 3, 5, 6, 8, 9, 14, 15)$ Design the control logic by implementing the Boolean function using a single 8×1 multiplexer, selecting appropriate select lines and input connections.	3
	(b)	Design a full subtractor using decoder.	2